MILITARY STANDARD

CONNECTIONS, ELECTRICAL
SOLDERLESS WRAPPED

TO ALL HOLDERS OF MIL-STD-1130B:

1. THE FOLLOWING PAGE OF MIL-STD-1130B HAS BEEN REVISED AND SUPERSEDES THE PAGE LISTED:

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
15	19 July 1979	15	12 December 1978
16	12 December 1978	(REPRINTED WITHOUT CHANGE)	

2. RETAIN THIS NOTICE AND INSERT BEFORE TABLE OF CONTENTS.

3. Holders of MIL-STD-1130B will verify that page changes and additions indicated above have been entered. This notice page will be retained as a check sheet. This issuance, together with appended pages, is a separate publication. Each notice is to be retained by stocking points until the Military Standard is completely revised or canceled.

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(Project 5935-3134)

FSC 5935

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5.6.1 Strip force. A completed solderless wrapped connection with the minimum number of turns listed in table II, shall be capable of meeting the minimum strip force specified in table III. The minimum strip force shall be obtained before the wrapped connection has been moved along the wrappost a length equivalent to the diameter of the conductor used on the wrappost.

TABLE III. Strip-force limits.

Wire Size	Conductor Diameter in Inches	(mm)	Minimum Strip Force (pounds) 0.025 Square	0.045 Square 0.06 x 0.03
30	0.010	(0.25)	2.0	-
28	0.0126	(0.32)	3.0	-
26	0.0159	(0.40)	4.0	6.0
24	0.0201	(0.51)	5.0	7.0
22	0.0253	(0.64)	-	8.0
20	0.0320	(0.81)	-	9.0
18	0.043	(1.02)	-	15.0

The samples shall be tested using a test fixture illustrated in figure 13. The clearance between the wrappost and the test fixture jaws shall not exceed 70 percent of the wire diameter. The stripping force shall be applied at a uniform rate of 1 to 10 inches per minute.

5.6.2 Gas tight. 75 percent of the corners in contact with uninsulated wire, except for the first and last turn, shall be gas tight when tested as follows:

Darken the assembly with ammonium sulphide gas after wrappost is exposed to aqua regia fumes.

a. The wrappost shall be suspended and corked in a 16 x 150 millimeter (mm) test tube containing approximately 1 to 2 milliliters (ml) of aqua regia solution (1:1 concentrated hydrochloric and nitric acids). The solution must not touch the wrappost assemblies. The wrappost assemblies shall be exposed to the aqua regia fumes for 10 minutes.

b. The chemically attacked wrappost shall be transferred to another 16 x 150 mm test tube containing approximately 1 ml of concentrated ammonium sulphide solution. Suspend the wrappost connections in the test tube so that they will not touch the ammonium sulphide solution. Close the test tube with a cork or rubber stopper and leave the specimens exposed to this atmosphere until they turn dark. Copper and gold sulphide color will be developed on all areas exposed to vapor.

c. Dry and unwrap the connections with a tool which does not scratch the post. The gastight areas will appear in bright, sharp contrast with discolored area.

5.6.3 Unwrap test. Wire on wrappost shall be capable of being unwrapped without conductor breakage. (See figures 14 and 15).

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5.6.3.1 Unwrap test (conductor brittleness). Place the unwrapping tool over the terminal post and engage its leading edge between the wrap end and the next wrap turn. Rotate the unwrapping tool until all the wire has been transferred onto the tool. Remove the tool with the loose helical coil from the terminal post. Holding the insulated portion of the wire firmly, rotate the tool unwinding the wire. The unwrapped wire need not be perfectly straight; waves and permanent deformation in it are permissible.

5.6.4 Wrapper resistance. When measured as illustrated on figure 16, at the current specified in table IV, the voltage drop across the wrapped connection shall not exceed 4 millivolts.

TABLE IV. Wrapper resistance.

AWG wire size	Conductor Diameter in inches	(mm)	Current to be used to determine wrapper resistance (amperes)
30	0.010	(0.25)	1.0
28	0.0126	(0.32)	2.0
26	0.0159	(0.40)	2.4
24	0.0201	(0.51)	2.4
22	0.0253	(0.64)	2.4
20	0.0320	(0.81)	7.5
18	0.0403	(1.02)	7.5

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